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Rigless Retrieval of a Stuck ESP Isolation Sleeve Using Real-Time Robotic E-Line Fishing Technology: A Case Study

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Abstract

This paper presents a case study on the successful retrieval of a mechanically stuck ESP isolation sleeve installed across a Y-tool in an offshore producer well. Multiple attempts to recover the sleeve using conventional fishing methods were unsuccessful. The objective of this operation was to restore well production and accessibility using a rigless, E-line solution capable of applying controlled, bi-directional force to overcome the mechanical lock-up. The methodologies, tools, and custom mechanical solutions employed in this intervention are discussed, offering practical insights for overcoming similar challenges. The operation demonstrates how advanced robotic E-line fishing systems, combined with custom mechanical adaptations, can be deployed to resolve complex downhole challenges and serve as a reliable alternative to traditional heavy interventions.

An investigation following the failure to retrieve the isolation sleeve using conventional methods revealed a design flaw that caused the inner collet to move upward in unison with the outer collet when force was applied. This interaction prevented the shear pins from breaking, resulting in a mechanical lock-up. To overcome this, a custom-designed spring-activated sub (SAS) was incorporated below an electrically activated fishing tool (EAF) and deployed with an E-line Stroker—marking the first field application of this concept—to deliver a controlled downward force on the inner mandrel while simultaneously pulling at the fish neck. This differential loading enabled the shear pins to shear without engaging the upper no-go, allowing the collets to collapse and the sleeve to be retrieved without the risk of jamming. Furthermore, a potential hydraulic lock scenario was identified. Unlike jars, which deliver a sudden, short-duration impact, the E-line Stroker and spring activated sub applies a gradual and consistent force. This controlled application of force is more effective in bleeding off trapped pressure and overcoming the hydraulic lock.

The isolation sleeve was successfully recovered in a single run. The EAF tool was able to latch onto the internal profile of the sleeve, which was confirmed using the surface read out measurement from the tool. The E-line Stroker was used to compress the SAS sub and provided the necessary force to shear the screws. Real-time data from the E-line stroker confirmed the stroke displacement and shear pin shearing, facilitating the collapse of the outer collets and subsequent release of the sleeve. Up to 35,000lbs of force were required to release the isolation sleeve. This confirmed the effectiveness of the spring actuated downward force

mechanism in overcoming the collet interlock. The spring active sub (SAS) in combination with the EAF tool enabled us to provide a separation force against the sleeve release mandrel at the same time anchoring with the EAF grapples to the main latch sleeve.

The successful retrieval of the isolation sleeve avoided a costly workover, resulting in substantial financial savings and a significant reduction in carbon emissions. This case demonstrates that robotic E-line fishing systems—equipped with real-time diagnostic capabilities—are not only effective but often superior to traditional fishing methods, enabling the successful recovery of various downhole obstructions that might otherwise be permanently stuck in the well.

Introduction and Background

Electric Submersible Pumps (ESPs) are widely used in artificial lift applications, and isolation sleeves installed across Y-tools are integral components that allow selective isolation of ESPs. These sleeves are typically retrieved using conventional intervention methods such as slickline or coiled tubing tools. However, mechanical complexity, restricted access, and component tolerances can pose substantial challenges during retrieval, especially in offshore wells where rig-based workovers are costly and logistically intensive.

In Well X, the operator faced such a challenge. An ESP isolation sleeve had been set across the Y-tool and had functioned normally until a scheduled intervention required its removal. Despite multiple recovery attempts using conventional fishing methods, the sleeve remained stuck. Analysis revealed a unique mechanical locking phenomenon exacerbated by an inherent design flaw.

Recognizing the limitations of conventional fishing methods, the operator opted for a fully rigless, robotic E-line fishing solution—an innovative approach made possible by recent advancements in e-line robotic intervention technology. These technologies offer precise control and real-time monitoring capabilities, significantly enhancing the efficiency and effectiveness of downhole fishing operations. This paper outlines the diagnostic findings, engineering innovations, execution methodology, and ultimate success of this first-of-its-kind intervention.

Additionally, the methodology outlined in this paper can serve as a guide for future similar fishing challenges, providing a structured approach to tool selection, parameter optimization, and risk mitigation for enhanced operational efficiency and success.

Wellbore Configuration and Constraints

The intervention was carried out in an offshore vertical well with the following parameters:

- **Depth:** 7995 ft
- **Bottomhole Temperature (BHT):** 200°F
- **Bottomhole Pressure (BHP):** 2900 psi
- **Fluid at fish depth:** Oil
- **H₂S/CO₂ presence:** 0% H₂S, 4% CO₂

Problem Statement

Upon investigation by the isolation sleeve supplier, it was determined that the isolation sleeve's dual collet design created an unintended mechanical interlock. The collets—meant to collapse upon shearing of internal pins—relied on axial force to release. However, when the sleeve was pulled against the upper no-go, the inner collet simultaneously travelled with the outer collet due to the lack of mechanical restraint between

the inner collet and the mandrel. This movement resulted in the inner collet pushing the outer collet deeper into the no-go profile, leading to a jammed assembly.

Because of this, the shear pins could not experience pure shear load and remained intact despite multiple surface jarring attempts. Without a method to isolate forces between the inner and outer mandrels—particularly to push the inner mandrel down while simultaneously pulling on the fish neck—the mechanical bind could not be broken.

Traditional fishing tools are limited to uni-directional force application. A novel approach using bi-directional force delivery in a controlled, real-time environment became the only feasible solution.

Job Planning and Method Selection

The first phase of job planning involved meetings with all stakeholders to gather relevant data and define objectives. The objectives were

- Engineering a rigless solution capable of recovering a jammed isolation sleeve that resisted conventional recovery techniques
- Implementing a fishing strategy that applied simultaneous upward and downward force vectors to create differential loading on the sleeve.
- Ensure real-time monitoring and precise execution, avoiding further damage to the completion and ensuring the Y-tool and tubing integrity remained uncompromised.
- Ultimate objective: Prevent a workover

To meet these, the selected tools and the design approach had to address two critical mechanical behaviours:

- The *inability to separate the sleeve collets* due to axial coupling between inner and outer mandrels.
- The *need for differential loading*—simultaneous downward force on the inner mandrel while pulling upward on the sleeve's main body.

This ruled out traditional fishing methods, which would lack the required precise control.

Technology Selection

Electric release pulling tool with SAS sub

Design and Integration of the Spring Activated Sub (SAS) with Electric Release Pulling tool. The Spring Activated Sub was at the heart of this intervention strategy. It was designed specifically to resolve the unique flaw in the sleeve's inner and outer collets mechanism. The SAS's design went through rapid but rigorous iterations based on failure analysis and system integration testing.

Key Functional Requirements of SAS.

- Deliver isolated downward force directly onto the sleeve's inner mandrel
- Function passively within the toolstring, without need for external power or triggering
- Control force delivery magnitude and sequence

Engineering Highlights.

- *Spring Stack:* Selected based on required force to shear the inner mandrel screws. Compression calibration was done during SIT to validate repeatable shear.

- *Spring Compression length*: Designed with 27 mm compression distance to match the elastic range and energy requirement to induce full shearing of pins.

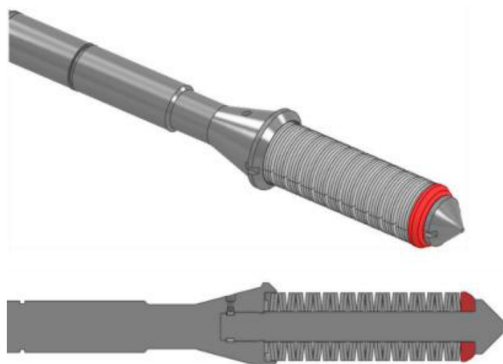


Figure 1—SAS pusher and prong design

The SAS sub was then connected to an **Electric release pulling tool** which serves as the interface to the isolation via its 3.5" internal fish neck.

The electric release fishing tool is designed to enhance and simplify fishing operations by enabling reliable disconnection of fishing strings or repositioning of the fish in the well. The tool's versatility allows for easy assembly of the required grapple type, either external or internal, by changing just a few parts. The surface read out functionality also enables ID/OD measurements of the profile being latched or grabbed onto providing confidence that the correct section of the fish is engaged. **Electrical disconnect** can be activated whenever necessary, and the tool can latch, release, and re-engage without the need to trip out of the hole to reset. The non-reliance on shear pins to disengage from the fish is a key feature of this tool, preventing premature release and enabling safe working loads up to 50% higher than standard pulling tools.

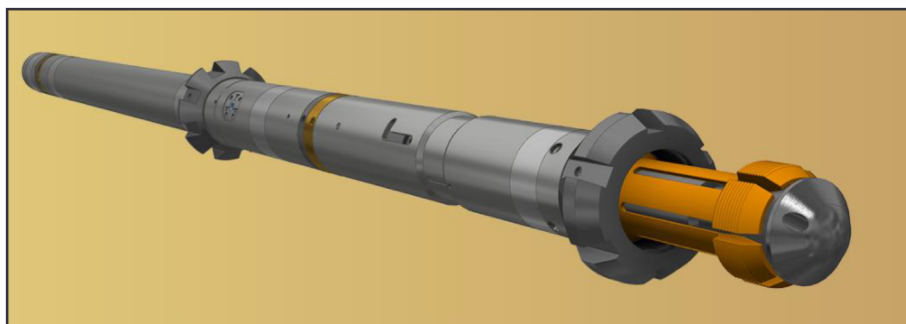


Figure 2—The Electric Release Fishing tool equipped with Internal Spear latch.

The tool features retractable keys or slips that facilitate engagement with a catch profile. In case of communication loss with the Electric Release Fishing Tool, the tool string can be fully recovered through the remote activation of a fail-safe timer release device. Additionally, the tool can effectively work with bottom hole assemblies that are difficult to catch onto a profile, minimizing the risk of premature release.

E-line Stroker

The E-line hydraulic stroker is designed to deliver precise axial force at specific depths downhole, regardless of the well's complexity. Unlike traditional methods like slickline, braided line, coiled tubing, and drill pipe, it offers exceptional control, directing force accurately to the required location.

When activated, the hydraulic stroker utilizes multiple anchoring points to distribute force evenly against the completion's wall, ensuring the tool is securely anchored during interventions. Once anchored, the

hydraulic piston provides a bidirectional push and pull force of up to 100,000 lbf. It features a fail-safe system to retract the anchors in case of power loss, ensuring safe recovery of the tool string and enhancing intervention reliability.

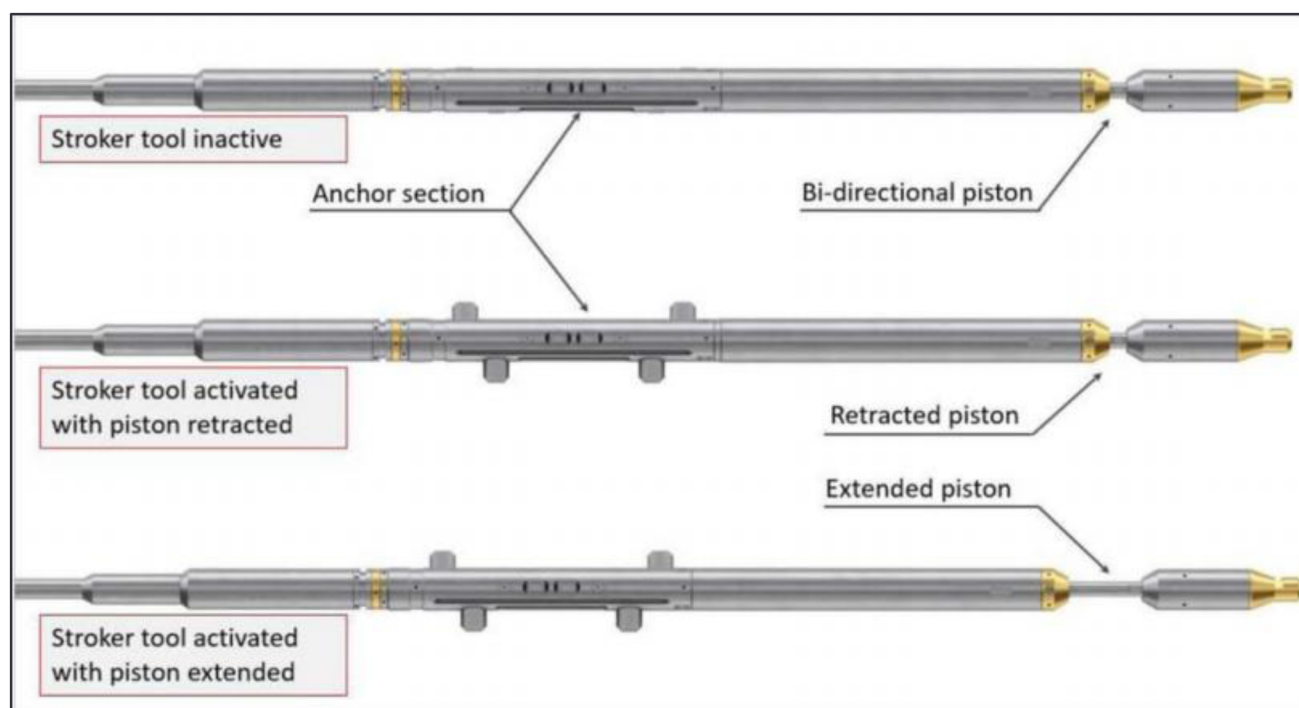


Figure 3—E-line hydraulic stoker in active and inactive modes

The tool allows precise control over the applied force, adjustable in 1500 lbf increments within a range of 10,000 lbf to 100,000 lbf. It can perform stroking without anchoring, improving control over the piston's position before engaging with the target. The hydraulic piston can be configured with various fixtures like pulling tools, shifting tools, and blind boxes.

Electric Release Device

The Electric Release Device is a tool that brought tremendous versatility and risk reduction to the operation. It provides one or more, predetermined locations where the toolstring can be parted should it become stuck.

The device can be activated via surface control or will release based on preset timing if power to the tool is lost. Leaving behind a standard 1-3/8-in. fishing neck reduces risk and ensures efficient fishing operations in the event of unplanned sticking.

System Integration Test

Prior to field deployment, extensive testing was carried out at the workshop. A representative isolation sleeve and Y-tool profile were used to:

- Validate tool compatibility.
- Confirm shear pin response to the calculated spring force.
- Test repeatability of the SAS activation and reset.
- Ensure correct sequence of motorized latching of the grapples and spring release.

Spring stack test

The spring stack was tested using a hydraulic cylinder. The spring stack was compressed and the measured travel was observed to be 27mm for completely compressing the springs. During the compression, three points (spring force Vs compression) were captured to determine the spring characteristics.

Operational Sequence.

- RIH to target depth and tag fishneck.
- Confirm fishneck engagement via surface read out feedback.
- Apply downward stroke (10,000 lbs) to compress SAS.
- Switch stoker to STATIC mode to release force.
- This will relieve tension and the spring decompresses and returns to its original shape. During this decompression, the stored energy is released and will be exerted on the shear pins. This will break the shear pins
- Shear pins rupture.
- Apply upward force incrementally (up to 35,000 lbs).
- Collets collapse and sleeve separates.
- Retrieve entire assembly to surface.

Execution

The toolstring was assembled and inserted into the test setup which comprised of a joint of 4.5" tubing connected to the Y tool assembly. The Isolation sleeve was installed inside the Y tool assembly prior to the test.

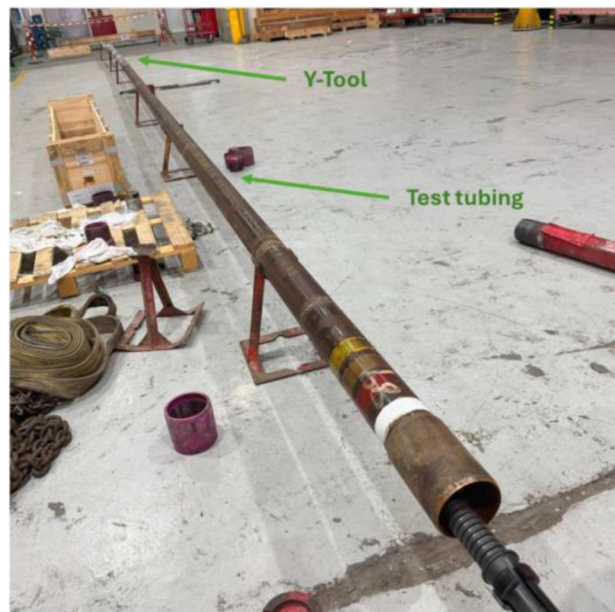


Figure 4—Test Setup

The tools string was then pushed into the Y tool assembly until the bottom of the tool tagged the inner shear out of the Isolation sleeve. Using a test camera, both landing of the SAS prong and the release sequence were visually monitored. The SAS Prong landed and centralized fully when pushing the tool string against

the Isolation Sleeve. During activation of the E-line Stroker to stroke down, compression of the springs was confirmed both visually and by surface readout from the E-line Stroker Piston travel.

After stroking down to compress the springs, the Electric Release pulling tool was activated to extend the grapples to engage with the upper fish neck of the Isolation Sleeve. Grapple extension readings from the software was confirmed, indicating that the Electric Release pulling tool grapples are in the correct position. Note: Monitoring the SRO readings from the Electric release pulling tool will provide visibility on the latch ID the grapples have engaged to.

To shear the sleeve's shear screws, the stroker was switched to "static" mode, effectively removing downward force. This allowed the compressed SAS springs to decompress and release their stored energy downward onto the inner sleeve mandrel. This decompression event applied the necessary shearing force to break the shear screws—without requiring active pulling from the stroker.

It was confirmed that the piston moved up by the SAS springs decompression when the stroke down force was removed. This was confirmed by monitoring the piston movement on the software as well as visually confirming with the test camera.

After confirming that the SAS springs have sheared the shear screws, a stroke-up force was applied with the E-line Stroker. During the stroke-up a loud noise was heard, indicating that the remainder of the SAS springs travel occurred. The system tests confirm that the stroke-up from the E-line Stroker ensures that the Isolation Sleeve fully opens to release position. Using low force, the E-line Stroker successfully pulled the Isolation Sleeve out of the Y-Tool nipple profile. After completely pulling out, the Isolation sleeve was inspected and confirmed to be in fully released position. The retrieved Isolation Sleeve is shown in [Figure 5](#).



Figure 5—Retrieved sleeve

Results and Observations. The test achieved complete retrieval of the isolation sleeve in a single run. Critical milestones and observations:

- **Successful Engagement:** The Electric Release pulling tool confirmed secure engagement through SRO feedback. Data confirmed the latch at the expected latch area.
- **Controlled Shearing:** The decompression of the SAS was smooth, and surface readouts indicated successful shearing of the pins. The movement of the Stroker piston aligned with predicted spring release behavior.
- **Post-Job Inspection:** The isolation sleeve showed full separation of inner and outer mandrels with all shear pins cleanly sheared. No damage was observed on the fishneck or Y-tool body.

The success of the fishing run validated the tool design, mechanical theory, and execution protocol.

Job Execution

Once the fishing program was finalized and a consensus was reached among all stakeholders regarding the overall fishing strategy, the equipment underwent a functionality check at the base before being swiftly mobilized offshore.

The toolstring was function-checked on the surface, and the system was verified in all operational modes before being run in the hole (RIH). As the toolstring was lowered, a correlation pass was conducted to confirm depth alignment. Tools were run in hole (RIH) to 5000 ft, and all toolstring components (E-line Stroker and Electric Release pulling tool) were function tested and passed.

First tagging attempt: The toolstring tagged the top of the fish (TOF) at 7,995.5 ft. With a 7 ft slack-off, the stroker anchors were set, and a 5,000 lbf downward stroke was applied to compress the SAS spring. However, no stroker piston movement was observed, indicating little to no spring compression. To verify engagement, the Electric Release pulling tool grapple was extended to its full stroke while monitoring SRO readings, aiming to confirm whether the spring had compressed and the grapple had reached the proper latch point. It was observed that the grapple had fully extended without any indication of engagement, confirming that the spring had not compressed and the grapple remained outside the fish. The grapple was then retracted, and a decision was made to re-tag.

Retagging: The toolstring was picked up and the fish was retagged and 10 ft of slack given, stroker anchors were set, and downward stroke was attempted but this time applying 10,000 lbs. Again, no stroker piston movement was recorded indicating the spring has not been compressed.

Overcoming decentralization: During troubleshooting, it was identified that the likely reason for the failure to compress the spring was the decentralization of the SAS sub. To address this, the toolstring was pulled up by 40 ft and then re-tagged. Upon re-tagging, the wireline was pulled up by 0.5 ft, and the stroker anchors were set at this depth. The rationale was that positioning the SAS sub 0.5 ft above the TOF during anchor setting would allow it to centralize as the stroker anchors naturally centralize the toolstring. Once anchored, a downward stroke force was applied, resulting in a 257 mm stroker piston movement—confirming successful compression of the SAS sub spring.

Once the spring has been compressed the Electric release pulling tool grapples were extended. Once the grapples were firmly latched onto the fish neck, the stroker downward force was removed and the stroker piston position retracted back from 256 mm to 237 mm, equivalent to a 19 mm back displacement in stroker piston position suggesting successful release of the spring stack and breaking of the isolation sleeve shear screws.

Then the stroker was activated in stroke up mode with 15,000 lbs force where the stroker piston was observed to move from 232 to 167 mm position (65 mm of travel).

- Stroker force was then increased to 40,000 lb, then the stroker piston was able to completely retract to 0 position at 34,500 lb tension indicating free up movement of the fish (this indicates that the stroker piston required force up to 34,500 lb to be able to fully free the sleeve).
- Stroker anchors were then released to extend the stroker piston and prepare the stroker to stroke up again.
- Another stroke up with 25,000 lb force was performed, this time the stroker piston moved the entire 360 mm (at 17,000 lb of tension) indicating the fish was free to move at this stage.
- As wireline started to pick up the tool string, the Wireline winch operator reported an increase of about 150 lbs in the weight of the tool string.
- Decision was made to pull out of hole (POOH), while keeping the grapple extended (engaged with the fish)
- Once on surface it was found that the fish has been successfully recovered.

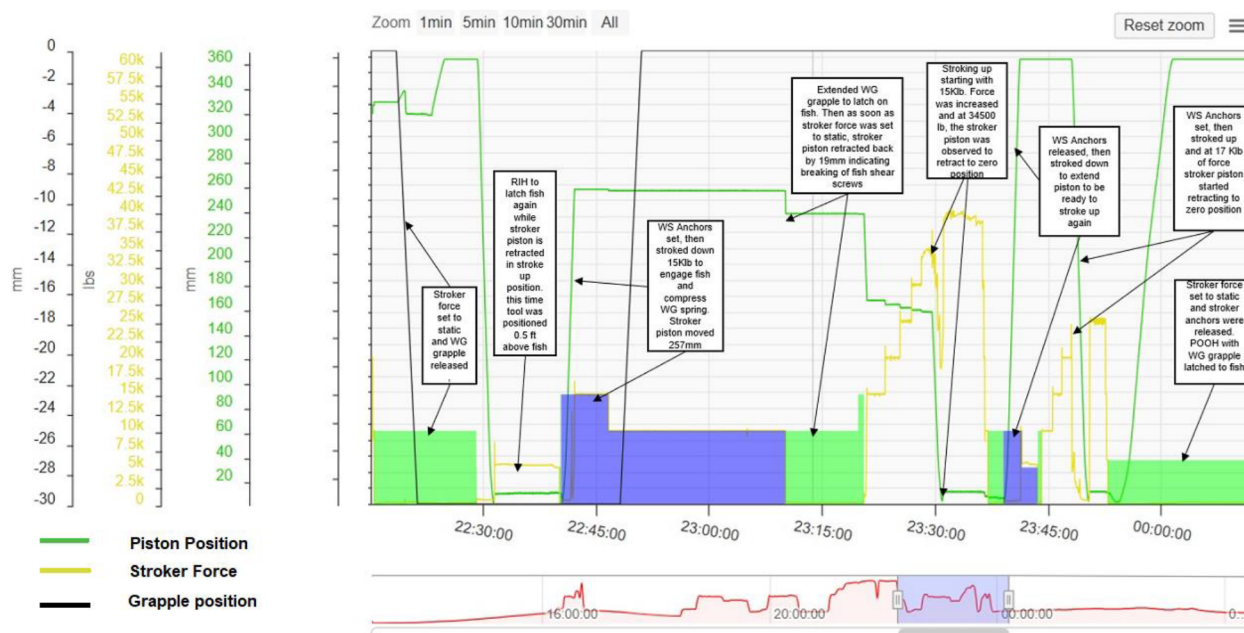


Figure 6—SRO data from Stroker and Electric release pulling tool confirming successful release of the Isolation sleeve

Conclusion & Lessons learned

The retrieval of the mechanically stuck isolation sleeve in Well X using a robotic E-line deployed toolstring represents a significant advancement in the domain of rigless ESP well interventions. The successful application of a custom-engineered Spring Activated Sub (SAS), integrated with the E-line Stroker and Electric Release Pulling tool, demonstrated that downhole mechanical challenges—previously deemed irretrievable via conventional fishing methods—can now be addressed with precision, control, and efficiency.

This field-first operation highlighted the ability to deliver bi-directional force in a controlled environment, simulate and validate tool behaviour through System Integration Testing, and execute a seamless real-time downhole sequence. The retrieval was achieved in a single run without compromising the integrity of the completion string or requiring a workover rig, underscoring the operational, economic, and environmental value of robotic E-line solutions.

The application of the SAS enabled a new class of mechanical intervention; spring-assisted downward compression targeting inner sleeve components; while the E-line Stroker provided high-precision force application and live monitoring. This intervention not only resolved a critical well access issue but also set a new benchmark for what is achievable using modular, programmable, and rigless intervention systems.

Lessons Learned

Pre-Job SIT adds predictability: The success of the operation was closely tied to the exhaustive System Integration Testing. The ability to simulate tool behavior under realistic conditions validated spring forces, confirmed shear values, and allowed refinements to the intervention procedure before going downhole.

Real-Time monitoring enhances decision-making: The Eline Stroker's ability to provide live force and displacement feedback allowed the team to track tool behavior with confidence, avoiding overpull or blind escalation of forces. It also provided verification of successful spring decompression and shear pin rupture.

Overcoming decentralization issues: Initial trials to compress the spring stack were unsuccessful due to potential misalignment. However, by introducing a 0.5 ft gap above TOF before anchoring the E-line Stroker improved tool alignment and enabled successful spring compression and shear screw breaking.

Collaboration across stakeholders is crucial: The operation's success was enabled by seamless collaboration between the end user, intervention tool provider engineering and operations teams, and the

Isolation sleeve manufacturer. Open technical dialogue and responsiveness were key to achieving first-run success.

Nomenclature

CT	Coiled tubing
E-line	Electric line
SL	Slick Line
LIB	Lead Impression Block
ID	Inner diameter
WOB	Weight on bit
MMC	Metal-matrix compound
SAS	Spring Activated Sub
OD	Outer Diameter
POOH	Pulled out of hole
TOF	Top of Fish
WD	Water Depth
MSL	Main Sea Level
SRO	Surface Read Out
RIH	Run in hole
SIT	System integration test

References

- Mangione, A., et al 2020. Significant Time Reduction and Cost Savings Realized Through the Use of Wireline Deployed Milling Services, to Enlarge a Tubing Hanger. Presented at the SPE/ICoTA Well Intervention Conference and Exhibition, The Woodlands, Texas, USA, 24–25 March. SPE-199845-MS. <https://doi.org/10.2118/199845-MS>.
- Ragaglia, S., et al 2021. Innovative Use of Wireline Tools Enables Successful Re-Entry on Subsea Wells. Presented at the Abu Dhabi International Petroleum Exhibition and Conference (ADIPEC), Abu Dhabi, UAE, 15–18 November. SPE-207442-MS. <https://doi.org/10.2118/207442-MS>.
- Dong, Y. 2022. E-line Milling of Isolation Valve Restores Well Access in the North Sea. Presented at the SPE Norway Subsurface Conference, Bergen, Norway, 26–27 April. SPE-209533-MS. <https://doi.org/10.2118/209533-MS>.
- Aattray, R., Annandale, C., and Grassick, D. 2008. Milling of Cast-Iron Bridge Plug with Electric-Line-Conveyed Technology. Presented at the SPE/ICoTA Coiled Tubing and Well Intervention Conference and Exhibition, The Woodlands, Texas, USA, 1–2 April. SPE-113288-MS. <https://doi.org/10.2118/113288-MS>.
- Stragiotti, L., and Karlsen, T. 2009. Successful Milling and Removal of a Permanent Bridge Plug with Electric-Line Tractor Conveyed Technology. Presented at the SPE/ICoTA Coiled Tubing and Well Intervention Conference and Exhibition, The Woodlands, Texas, USA, 31 March–1 April. SPE-121539-MS. <https://doi.org/10.2118/121539-MS>.